Equivalent Modeling of Microbump Layer in Microsystem for Thermal Analysis Based on Differential Idea

Kui Li, Xin Wang, Zhixiang Zhang, Nailiang Kuang, Yujun Yang, Jing Li, and Wu Gao

***Abstract*— The through-silicon-vias (TSV) packaging structure in the microsystem is complex, especially there are a large number of microbumps. Meanwhile, the microsystem structure size has a cross-scale problem. Then, detailed modeling will bring difficulties in meshing and simulation calculation efficiency. Based on the actual bump structure in the microbump layer, a calculating method for the anisotropy equivalent thermal conductivity of the microbump layer is proposed by using differential ideas combined with the heat transfer theory. The thermal simulation analysis of the detailed model and the equivalent model is completed. The comparison simulation results show that the equivalent modeling method has high accuracy and greatly calculation efficiency. By comparing the thermal test result, the simulation result is basically consistent with the test result, which verified the accuracy of the equivalent modeling method again. Later, the influence of the bump radius, the bump pitch, and the microbump layer height on the equivalent thermal conductivity of the microbump layer is mainly studied, which provides theoretical support on thermal design for the microbump layer in the microsystem.**

***Index Terms*— Differential idea, equivalent modeling, microbump layer, microsystem, thermal analysis.**

NOMENCLATURE

R	Bump radius, mm.
p	Bump pitch, mm.
h	Height of the microbump layer, mm.
z	Z-coordinate position of the microlayer structure, mm.
r_z	Diameter of bump in the microlayer structure, mm.
Q	Heat flow power, W.
A	Cross-sectional area of the conductor in the direction of heat flow, mm^2.
ΔT	Temperature difference between the upper and lower surfaces, °C.
k	Thermal conductivity of the material, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
R_z	Thermal resistance of the microlayer structure in the Z-direction, $K{\cdot}W^{-1}$.
R_u	Thermal resistance of the underfill, $K{\cdot}W^{-1}$.
R_b	Thermal resistance of the bump, $K{\cdot}W^{-1}$.
k_z	Z-direction equivalent thermal conductivity of the microlayer structure, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
k_u	Thermal conductivity of the underfill, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
k_b	Thermal conductivity of the bump, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
dz	Thickness of the microlayer structure, mm.
A_z	Cross-sectional area of the microlayer structure in the vertical Z-direction, mm^2.
A_u	Cross-sectional area of the underfill in the vertical Z-direction, mm^2.
A_b	Cross-sectional area of the bump in the vertical Z-direction, mm^2.
R_Z	Thermal resistance of the microbump layer unit cell in the Z-direction, $K{\cdot}W^{-1}$.
R_{zi}	Thermal resistance of the i-layer microlayer structure in the Z-direction, $K{\cdot}W^{-1}$.
dy	Height of a microsquare body, mm.
y	Y-coordinate position of the microsquare body, mm.
r_z	Radius of the bump of the microlayer structure, mm.
a_y	Length of the bump material in the microsquare body, mm.
k_y	Equivalent thermal conductivity in the Y-direction of the microsquare body, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
R_{By}	Thermal resistance of part B in the Y-direction, $K{\cdot}W^{-1}$.
R_{yi}	Thermal resistance of the i-layer microsquare body in the Y-direction, $K{\cdot}W^{-1}$.
k_{Ay}	Part A thermal conductivity of the microlayer structure, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
R_y	Series thermal resistance of microlayer structure in the Y-direction, $K{\cdot}W^{-1}$.
R_Y	Thermal resistance of the microbump layer unit cell in the Y-direction, $K{\cdot}W^{-1}$.
R_{yi}	Thermal resistance of the i-layer microlayer structure in the Y-direction, $K{\cdot}W^{-1}$.
k_Y	Equivalent thermal conductivity of the microbump layer unit cell in the Y-direction, $W{\cdot}m^{-1}{\cdot}K^{-1}$.

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M	Total mass of the microbump layer unit cell, kg.
C	Equivalent specific heat capacity, $J{\cdot}kg^{-1}{\cdot}k^{-1}$.
m_1	Mass of the bump, kg.
C_1	Specific heat capacity of the bump, $J{\cdot}kg^{-1}{\cdot}k^{-1}$.
m_2	Mass of the underfill, kg.
C_2	Specific heat capacity of the underfill, $J{\cdot}kg^{-1}{\cdot}k^{-1}$.
k_s	Thermal conductivity of the solid material, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
T_s	Temperature of the solid material, °C.
S	Source term due to the heat generation of the chip, $W{\cdot}m^{-3}$.
T_b	Temperature at the interfaces of the bump, °C.
T_s	Temperature at the interfaces of the underfill, °C.
ρ_f	Density of the air, $kg{\cdot}m^{-3}$.
c_f	Specific heat of the air, $J{\cdot}kg^{-1}{\cdot}k^{-1}$.
μ_f	Viscosity of the air, $kg{\cdot}m^{-1}{\cdot}s^{-1}$.
β_f	Volumetric thermal expansion coefficient of the air, $°C^{-1}$.
k_f	Conductive coefficient of the air, $W{\cdot}m^{-1}{\cdot}K^{-1}$.
u	Velocity of the airflow in the x-direction, $m{\cdot}s^{-1}$.
v	Velocity of the airflow in the y-direction, $m{\cdot}s^{-1}$.
w	Velocity of the airflow in the z-direction, $m{\cdot}s^{-1}$.
P	Pressure of the air, *Pa*.
g	Gravitational acceleration, $m{\cdot}s^{-2}$.
T_f	Temperature of the air, °C.
T_∞	Temperature of the ambient, °C.
int	Interface of the solid material and the airflow.
$\boldsymbol{n}$	Normal unit vector to the interface.
s	Solid material.
f	Airflow.

I. Introduction

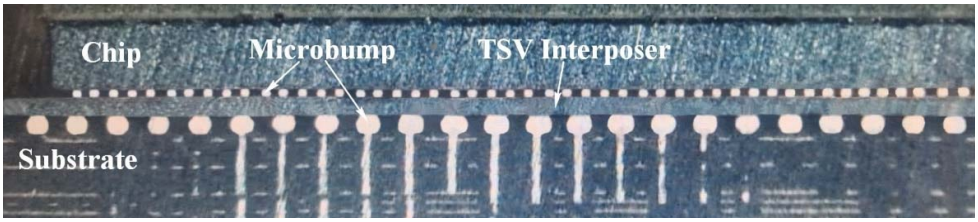
With the development of microelectronics along the road of Moore's law, microsystems based on 3-D integration technology have entered a new development stage. The 3-D integration technology with through-silicon-vias (TSV) has been widely regarded as the leading technology in the field of high-density packaging in the future and an effective way to break Moore's law [1]–[6]. Compared with the traditional 2-D packaging, 3-D integration technology based on TSV adopts the vertical interconnection way to realize a high-density packaging for chip stacking [7]–[9]. The increases of the power density in highly integrated microsystem and stacked chips number result in a series of thermal management problems [10]–[12], which directly affect the life reliability of the product and attract great attentions. Numerical simulation technology can greatly shorten the product development cycle, reduce design cost, and improve thermal reliability of products and has become an indispensable means in the product thermal management design and the package thermal characteristics evaluation [13]–[17].

For the microsystem with complex TSV package structure, detailed modeling will bring difficulties in mesh generation and low computational efficiency due to the large number of TSVs and microbumps, as well as the cross-scale problem caused by the large size difference of the structure. These problems will be exacerbated especially when the module is used for system simulation analysis. Therefore, the equivalent modeling of complex packages can greatly reduce computational cost, bringing great engineering application values.

A large number of studies have been carried out on thermal equivalent modeling of TSV structures by numerical calculation or simulation methods. Chengdi *et al.* [18] proposed a calculation method for equivalent thermal conductivity of TSV interposers based on finite element analysis, which was verified by numerical simulation, and analyzed the temperature distribution effect on the diameter and pitch of TSVs and the thickness of SiO_2. Ningyu *et al.* [19] applied the full-scale numerical simulation method based on the equivalent thermal conductivity model to the simulation modeling of fan-out packages, and the research showed that it ensured the high modeling accuracy and calculation efficiency. Fangwei *et al.* [20] and Wenchao and Hao [21] selected TSV structural unit to deduce the equivalent anisotropic thermal conductivity by numerical calculation method and applied it to 3-D packaging simulation, which obtained good application results. Chien *et al.* [22] obtained the temperature value through detailed modeling simulation for a TSV unit, then calculated the equivalent thermal conductivity by Fourier law, and determined the empirical formula of equivalent thermal conductivity under different structural parameters for simulation calculation. Yudan *et al.* [23] achieved equivalent model using analytical thermal model and finite element analysis. John and Gong [24] completed the simulation analysis of TSV chips by numerical simulation modeling with Fourier law. The above researches mainly provided reference methods and means for equivalent modeling of TSV structure, but there were few equivalent studies focused on microbumps and underfill layers, which also affect the simulation calculation amount. Mingyi *et al.* [25] completed the equivalent calculation for bump–underfill by the numerical equivalent method, and Oprins *et al.* [26] calculated the equivalent in-plane and out-of-plane thermal conductivity for an underfilled and bump array based on a finite element model. In the above studies, solder bumps were simply equivalent to square columns without considering the actual structure, while the actual bumps were spherical with truncated crowns. There is a big difference between two structures, which will also bring equivalent calculation error for thermal parameters. The metallographic morphology of typical microsystem bumps profile is shown in Fig. 1.

In order to solve the modeling problem for the microbump layer composed of actual spherical truncated crown bumps and underfills, an equivalent calculation method of thermal parameters based on differential idea is proposed, and the parameters of the anisotropic thermal conductivity and the specific heat capacity are calculated. Then, an equivalent thermal simulation model of the microsystem is established, and the accuracy of the equivalent modeling is verified by comparing thermal transient test results. The equivalent calculating method in this article has higher accuracy through the comparison analysis with the reference method. The influence of the bump radius, the bump pitch, and the microbump layer

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(1) Microsystem package structure profile



(2) Local solder bumps appearance

Fig. 1. Metallographic morphology of typical microsystem bumps profile. (1) Microsystem package structure profile. (2) Local solder bumps appearance.

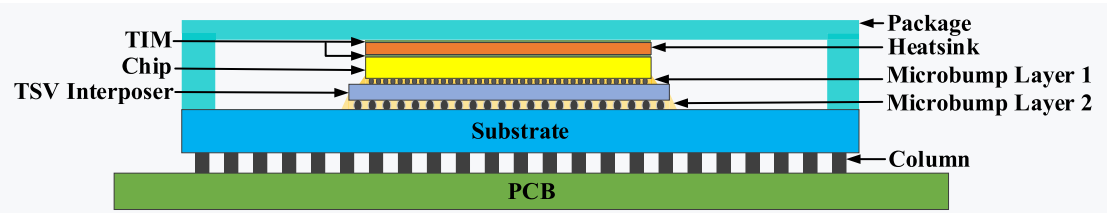


Fig. 2. Schematic of microsystem structure.

height on the heat dissipation characteristic for the microbump layer is analyzed, which provides reference for the microbump layer design in thermal management of microsystems.

II. Structure Description of Typical Microsystem

Fig. 2 describes the package structure of a typical microsystem, including chips, thermal interface materials (TIMs), heat sink, TSV interposer, microbump layer 1, microbump layer 2, package, substrate, and welding columns. At the bottom of the chip, the flip-chip process is adopted to welding microbumps on TSV interposer, forming the silicon component. The silicon component is welded on the substrate through bumps. In order to reduce the stress level of bumps, the bottom of the chip and the silicon component is, respectively, filled by underfills to form microbump layers. The microsystem structure includes two microbump layers. The chip is connected to the cover plate through TIM and the heat sink. The number of bumps in the module exceeds 10 000. Meanwhile, the minimum size in the module is the microbump diameter 0.08 mm, and the maximum size of the module is the microsystem side 43 mm. The ratio above is 1:537.5, which exceeds two orders of magnitude. Therefore, both the cross-scale problem in package structures and the large number of bumps will lead to difficulties in mesh generation and low computational efficiency in thermal simulation analysis for microsystems.

The microsystem structure shown in Fig. 2 is composed of two microbump layers. The microbump layer 1 is located below the chip, and the bump radius R is 0.045 mm, and the bump pitch p is 0.2 mm. The microbump layer 2 is located below the silicon interposer, and the bump radius R is 0.125 mm, and the bump pitch p is 0.5 mm. In the welding process, the solder ball and solder paste gradually melt into liquid together when the temperature reaches the solder's melting point. Then, the solder ball is compressed on account of the device's gravity. The solder ball solidifies and collapses with the temperature decreasing. The above process is called the welding collapse. According to the process experience, the height h of the microbump layer 1 and the microbump layer 2 is, respectively, 0.07 and 0.15 mm after the welding collapse for Sn63Pb37 bumps. The material parameters of the bump and the underfill in the microbump layer are shown in Table I.

III. Equivalent Calculation of Thermal Parameters for Microbump Layer

The microbump layer in the microsystem is a composite entity composed of bumps and underfills, as shown in Fig. 3. According to the heat conduction theory, the composite entity composed of different materials can be homogenized approximately equivalent from the perspective of macroscopic thermal diffusion effect, so the microbump layer can be equivalently treated by the homogenized entity. The principle of

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TABLE I
MATERIAL PARAMETERS OF THE MICROBUMP LAYER

Structure	Material	Thermal conductivity/$W{\cdot}m^{-1}{\cdot}k^{-1}$	Density/$kg{\cdot}m^{-3}$	Specific heat capacity/$J{\cdot}kg^{-1}{\cdot}k^{-1}$
Bump	Sn63Pb37	50.9	8400	150
Underfill	Adhesive	0.2	1120	1400

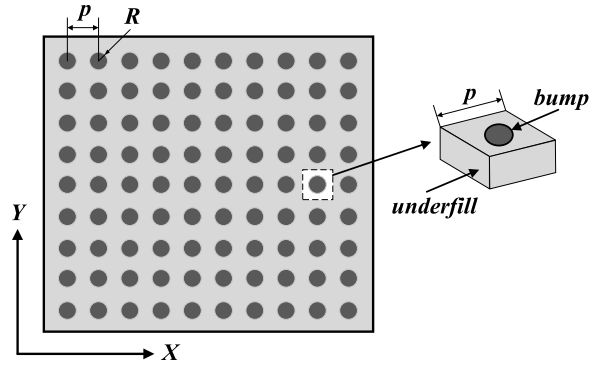


Fig. 3. Microbump layer structure diagram in microsystem.

homogeneous entity equivalence includes the same dimension and the same thermal conductivity before and after simplification. The key problem is how to calculate the effective thermal conductivity through a reasonable method, and then, a solid model is established to replace the microbump layer.

A. Equivalent Thermal Conductivity Calculation

In general, microbumps in microsystems are arranged in a regular array form, so a single bump and a part of underfill in the microbump layer are selected as the basic unit cell for theoretical calculation. The bump is a truncated crown ball and the underfill is a hexahedral square shape. The unit cell of the microbump layer is shown in Fig. 4. Since the microbump layer is not a completely homogeneous texture, the structure characteristics of the Z- and XY-directions for the microbump layer are different, and the heat dissipation characteristic of the microbump layer will also show anisotropic features, so equivalent thermal conductivities in the Z- and XY-directions should be calculated, respectively.

The bump is generally simplified to a square shape during equivalent calculation analysis, and some studies even directly take the diameter as the square shape side length through the maximum shape equivalent method in place of volume equivalence. While the actual bumps in the microbump layer are spherical with truncated crowns, so a big calculation error will be brought by the simplification method above due to the large difference between two structures. In this article, the equivalent thermal conductivity of the microbump layer is calculated theoretically with the differential idea. First, the equivalent thermal conductivity of the microlayer structure is calculated by means of the series and parallel theory of thermal resistances after a microlayer structure is extracted from the middle of the basic unit cell as the calculation object. Then, the equivalent thermal conductivity of the basic unit is obtained by integrations. The extraction schematic of the microlayer structure of basic unit in the microbump layer is shown in Fig. 4 (right).

1) Z-Direction Equivalent Thermal Conductivity Calculation: The thermal conductivity calculation of the basic unit in the microbump layer in the Z-direction can be seen in Fig. 5. According to the symmetry of the model, the upper part can be taken as the calculation object. The main view of the basic unit cell in the microbump layer is shown in Fig. 5 (left). As can be seen from the figure, the microlayer structure can be regarded as a regular cylinder with a very small height because of the very small thickness of the extracted microlayer structure. The cylinder and cross section are shown in Fig. 5 (right). In the figure, z is the Z-coordinate position of the microlayer structure, and r_z is the diameter of bump in the microlayer structure.

According to Fourier theory of heat conduction

$$Q = \frac{k \times A \times \Delta T}{h} \tag{1}$$

where Q is the heat flow power, A is the cross-sectional area of the conductor in the direction of heat flow, ΔT is the temperature difference between the upper and lower surfaces, h is the height of the conductor, and k is the thermal conductivity of the material. The corresponding thermal resistance calculation formula in situation of heat conduction is as follows:

$$R = \frac{h}{k \times A}. \tag{2}$$

As can be seen from the right figure of the calculation model in Fig. 5, the overall thermal resistance of the microlayer structure in the Z-direction can be considered as the thermal resistance in parallel between thermal resistances of the bump and the underfill. The thermal resistance parallel of the microlayer structure in the Z-direction can be expressed as

$$\frac{1}{R_z} = \frac{1}{R_u} + \frac{1}{R_b} \tag{3}$$

where R_z is the thermal resistance of the microlayer structure in the Z-direction, and R_u and R_b are the thermal resistances of the underfill and the bump, respectively.

Substituting (2) into (3), the thermal resistance in parallel can be expressed as

$$\frac{k_z \times A_z}{dz} = \frac{k_u \times A_u}{dz} + \frac{k_b \times A_b}{dz} \tag{4}$$

where k_z is the Z-direction equivalent thermal conductivity of the microlayer structure, k_u and k_b are the thermal conductivities of the underfill and the bump, respectively, dz is the thickness of the microlayer structure, A_z is the cross-sectional area of the microlayer structure in the vertical Z-direction, and

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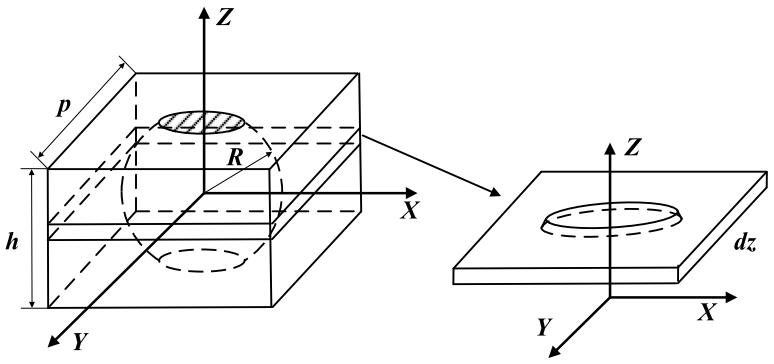


Fig. 4. Unit cell of the microbump layer and the microlayer structure extraction schematic.

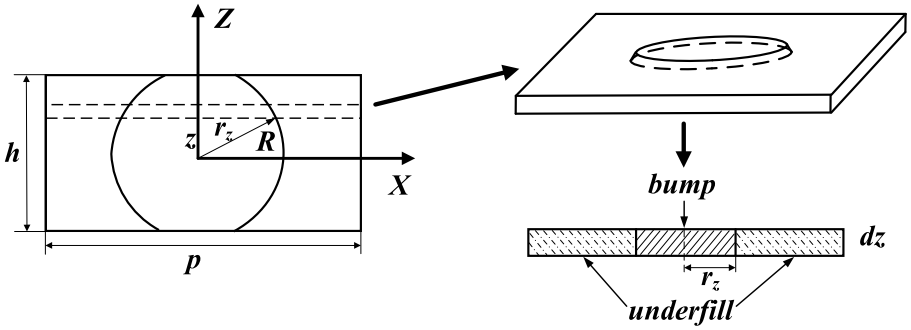


Fig. 5. Thermal conductivity calculation schematic for the unit cell of the microbump layer in the Z-direction.

A_u and A_b are the cross-sectional areas of the underfill and the bump in the vertical Z-direction, respectively.

The volume ratio between the bump and the underfill is

$$\alpha = \frac{V_b}{V_u} = \frac{A_b \times dz}{A_u \times dz} = \frac{A_b}{A_u}. \tag{5}$$

The cross-sectional areas of the bump and the underfill, respectively, are

$$A_b = \pi \cdot r_z^2 = \pi \cdot \left(R^2 - z^2\right) \tag{6}$$

$$A_u = p^2 - \pi \cdot \left(R^2 - z^2\right). \tag{7}$$

The relationship of cross-sectional areas between the microlayer structure, the underfill, and the bump is as follows:

$$A_z = A_u + A_b. \tag{8}$$

Substituting (5) and (6) into (4), the Z-direction equivalent thermal conductivity of the microlayer structure can be deduced as

$$k_z = \frac{k_u + \alpha \cdot k_b}{1 + \alpha}. \tag{9}$$

As can be seen from the left figure of the calculation model in Fig. 4, the overall thermal resistance of the microbump layer unit cell in the Z-direction can be considered as a series thermal resistance of countless microlayer structures. A series thermal resistance of the microbump layer unit cell in the Z-direction can be expressed as

$$R_Z = \sum_{i=0}^{\infty} R_{zi} \tag{10}$$

where R_Z is the thermal resistance of the microbump layer unit cell in the Z-direction, and R_{zi} is the thermal resistance of the i-layer microlayer structure in the Z-direction.

After substituting (2) into (10) and differential calculating instead of summation, a series thermal resistance can be expressed as

$$\frac{h/2}{k_Z \times A_z} = \int_0^{h/2} \frac{dz}{k_z \times A_z}. \tag{11}$$

Furthermore, k_Z is given as

$$k_Z = \frac{h}{2\int_0^{h/2} \frac{dz}{k_z}}. \tag{12}$$

Substituting (6), (7), and (9) into (12), the equivalent thermal conductivity of the microbump layer unit cell in the Z-direction can be expressed as

$$k_Z = \frac{h}{2\int_0^{h/2} \frac{p^2 dz}{\pi\left(R^2 - z^2\right)(k_b - k_u) + p^2 k_u}}. \tag{13}$$

2) X- or Y-Direction Equivalent Thermal Conductivity Calculation: From the top view of the extracted microlayer structure, the XY plane of the microlayer structure is shown in Fig. 6. According to the symmetry of the model, the upper part is chosen as the calculation object, and the equivalent calculation is carried out by taking the Y-direction thermal conductivity as an example. It can be seen from the figure that the microlayer structure in the XY plane can be divided into parts A and B. Part A is composed of underfills, and part

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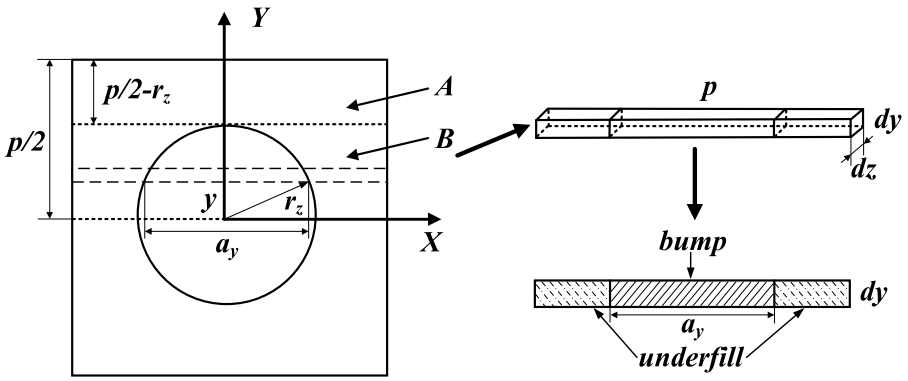


Fig. 6. *XY* plane model of the microlayer structure.

B is composed of bumps and underfills. The overall thermal resistance of the microlayer structure in the Y-direction can be considered as a series thermal resistance between A and B to form a series thermal conduction path.

The thermal resistance of part B is not simply composed of two materials in parallel because the bump is round. Similarly, the differential idea can also be adopted to extract the tiny middle section of part B for analysis. Due to its very small thickness, it can be regarded as a microsquare body with a height of dy, a length of p, and a depth of dz, as shown in Fig. 6. According to the parallel relation of thermal resistances of this section, the equivalent thermal conductivity in the Y-direction of this section is obtained, and then, the equivalent thermal conductivity in the Y-direction of part B is calculated out through the series relations of countless tiny square bodies. In Fig. 6, y is the Y-coordinate position of the microsquare body, r_z is the radius of the bump of the microlayer structure, and a_y is the length of the bump material in the microsquare body.

According to the parallel theory of thermal resistance calculation in the previous chapter, the equivalent thermal conductivity in the Y-direction of the microsquare body can be referred to (9). That is

$$k_y = \frac{k_u + \alpha \cdot k_b}{1 + \alpha}. \tag{14}$$

The cross-sectional areas of the bump and the underfill of the microsquare body are

$$A_b = dz \cdot a_y = dz \cdot 2\sqrt{\left(R^2 - y^2\right)} \tag{15}$$

$$A_u = dz \cdot \left(p - 2\sqrt{\left(R^2 - y^2\right)}\right). \tag{16}$$

Similarly, the overall thermal resistance of part B in the Y-direction is composed of numerous microsquare bodies in series, referred in (10)

$$R_{BY} = \sum_{i=0}^{\infty} R_{yi} \tag{17}$$

where R_{By} is the thermal resistance of part B in the Y-direction, and R_{yi} is the thermal resistance of the i-layer microsquare body in the Y-direction.

Substituting (2) into (17) and differential calculating instead of summation, a series thermal resistance can be expressed as

$$\frac{r_z}{k_{BY} \times A_{yi}} = \int_0^{r_z} \frac{dy}{k_y \times A_{yi}}. \tag{18}$$

Furthermore, k_{BY} is given as

$$k_{BY} = \frac{r_z}{\int_0^{r_z} \frac{dy}{k_y}}. \tag{19}$$

Substituting (14)–(18) into (19), the equivalent thermal resistance of part B in the Y-direction can be expressed as

$$k_{By} = \frac{r_z}{\int_0^{r_z} \frac{pdy}{2\sqrt{\left(R^2-y^2\right)}(k_b-k_u)+pk_u}}. \tag{20}$$

The part A thermal conductivity of the microlayer structure is the same as underfill

$$k_{AY} = k_u. \tag{21}$$

The series thermal resistance of microlayer structure in the Y-direction can be expressed as

$$R_y = R_{Ay} + R_{By}. \tag{22}$$

Substituting (2) into (22), that is

$$\frac{p/2}{k_y \times A_y} = \frac{p/2 - r_z}{k_{Ay} \times A_y} + \frac{r_z}{k_{By} \times A_y}. \tag{23}$$

Furthermore, the thermal conductivity of the microlayer structure in the Y-direction can be deduced as follows:

$$k_y = \frac{p/2 \cdot k_{By} k_u}{r_z\left(k_u - k_{By}\right) + p/2 \cdot k_{By}}. \tag{24}$$

As can be seen from the left figure of the calculation model in Fig. 4, the overall thermal resistance of the microbump layer unit cell in the Y-direction can be considered as the thermal resistance in parallel of countless microlayer structures. The thermal resistance in parallel of the microbump layer unit cell in the Y-direction can be expressed as

$$\frac{1}{R_Y} = \sum_{i=0}^{\infty} \frac{1}{R_{yi}} \tag{25}$$

where R_Y is the thermal resistance of the microbump layer unit cell in the Y-direction, and R_{yi} is the thermal resistance of the i-layer microlayer structure in the Y-direction.

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After substituting (2) into (25) and differential calculating instead of summation, the thermal resistance in parallel can be expressed as

$$\frac{k_Y \times A_Y}{p} = \int_0^{h/2} \frac{k_y \times A_{Yi}}{p} \tag{26}$$

$$\frac{k_Y \times p \times h/2}{p} = \int_0^{h/2} \frac{k_y \times p \times dz}{p}. \tag{27}$$

Furthermore, k_Y is given as

$$k_Y = \frac{2\int_0^{h/2} k_y dz}{h}. \tag{28}$$

Substituting (20) and (24) into (28), the equivalent thermal conductivity of the microbump layer unit cell in the Y-direction can be expressed as

$$k_Y = \int_0^{h/2} \frac{2p \cdot k_{By} \cdot k_u}{2r_z \cdot h \cdot (k_u - k_{By}) + p \cdot h \cdot k_{By}} dz. \tag{29}$$

B. Specific Heat Capacity Equivalent Calculation

The equivalent specific heat capacity parameters are calculated by mass ratio equivalent in the following:

$$C = \frac{m_1}{M} C_1 + \frac{m_2}{M} C_2 \tag{30}$$

where M is the total mass of the microbump layer unit cell, C is the equivalent specific heat capacity, m_1 is the mass of the bump, C_1 is the specific heat capacity of the bump, m_2 is the mass of the underfill, and C_2 is the specific heat capacity of the underfill.

C. Governing Equations and Boundary Conditions

To simply the numerical simulations, the following assumptions are made.

1) Steady state.
2) The airflow is incompressible.
3) The Boussinesq approximation is used to simulate the density variation of the air caused by heat and natural convection.
4) Other thermophysical properties of the air are constant.

For the stated assumptions above, the governing equations of the system are given as follows.

1) For the Solid: The steady-state heat conduction equation in the x-, y-, and z-directions can be given as

$$k_s \left(\frac{\partial^2 T_s}{\partial x^2} + \frac{\partial^2 T_s}{\partial y^2} + \frac{\partial^2 T_s}{\partial z^2} \right) + S = 0 \tag{31}$$

where k_s represents the thermal conductivity of the solid material, T_s represents the temperature of the solid material, and S is the source term due to the heat generation of the chip.

For the heat conduction between the bump and the underfill, the temperature and heat flux at the interfaces of them should be continuous and described as follows:

$$T_b = T_u \tag{32}$$

$$k_b \frac{\partial T}{\partial \mathbf{n}}\bigg|_{b,int} = k_u \frac{\partial T}{\partial \mathbf{n}}\bigg|_{u,int} \tag{33}$$

where the subscript "int" represents the interface of the bump and the underfill, and $\boldsymbol{n}$ is the normal unit vector to the interface. b and u represent the bump and underfill, respectively.

2) For Airflow: Continuity equation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0. \tag{34}$$

Momentum equation

$$\rho_f \left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z} \right) = \frac{\partial P}{\partial x} + \mu_f \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \tag{35}$$

$$\rho_f \left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + v\frac{\partial w}{\partial z} \right) = \frac{\partial P}{\partial y} + \mu_f \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho_f g \beta_f (T_f - T_\infty) \tag{36}$$

$$\rho_f \left(u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z} \right) = \frac{\partial P}{\partial z} + \mu_f \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right). \tag{37}$$

Energy equation

$$\rho_f c_f \left(u\frac{\partial T_f}{\partial x} + v\frac{\partial T_f}{\partial y} + w\frac{\partial T_f}{\partial z} \right) = k_f \left(\frac{\partial^2 T_f}{\partial x^2} + \frac{\partial^2 T_f}{\partial y^2} + \frac{\partial^2 T_f}{\partial z^2} \right) \tag{38}$$

where ρ_f, c_f, μ_f, β_f, and k_f are the density, specific heat, viscosity, volumetric thermal expansion coefficient, and conductive coefficient of the air, respectively, u, v, and w are the velocities of the airflow in the x-, y-, and z-directions, respectively, P is the pressure of the air, g is the gravitational acceleration, T_f is the temperature of the air, and T_∞ is the temperature of the ambient.

The temperatures of the ambient and the cold plate are considered constant as 25 °, and the power of heat generated by the chip is 10 W. For the numerical domain, no-slip boundary condition is applied on all the solid surfaces, and coupled condition is applied on the interface of the solid material and the airflow, which could be written as

$$k_s \left(\frac{\partial T_s}{\partial \mathbf{n}} \right)_{int} = k_f \left(\frac{\partial T_f}{\partial \mathbf{n}} \right)_{int}, \quad T_s = T_f \tag{39}$$

where the subscript "int" represents the interface of the solid material and the airflow, $\boldsymbol{n}$ is the normal unit vector to the interface, and s and f represent the solid material and the airflow, respectively.

IV. Thermal Simulation and Transient Test Verification

In order to verify the accuracy of the equivalent modeling method for the microbump layer, the equivalent parameters of the microbump layer in the typical microsystem structure, as shown in Fig. 2, are calculated, and then, the steady thermal result of the chip in the microsystem is simulated. Then, the actual heating process is monitored with T3ster thermal transient tester, and the modeling method is verified by comparing the test and simulation results.

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TABLE II
EQUIVALENT THERMAL CHARACTERISTIC PARAMETERS OF EACH MICROBUMP LAYERS

Microbump layer	$k/W{\cdot}m^{-1}{\cdot}K^{-1}$		$\rho/kg{\cdot}m^{-3}$	$C/J{\cdot}kg^{-1}{\cdot}k^{-1}$
	Z direction	*XY* direction		
Microbump layer 1	6.25	0.33	1562.96	1002.13
Microbump layer 2	8.82	0.37	2700.67	525.45

A. Thermal Simulation

1) Thermal Parameters Equivalent Calculation: The formulas of equivalent thermal conductivity in the above chapter are all complex integral functions, especially the thermal conductivity in the *XY*-direction needs to be integrated twice. It is very difficult to directly calculate the analytical formula, so that the parameters can be substituted into (13), (20), and (29), respectively, to calculate equivalent thermal conductivities. According to the above method, the parameters of microbump layers are substituted into the above formula to obtain the equivalent thermal parameters of the homogenization models for the microbump layers 1 and 2, as shown in Table II.

2) Thermal Simulation Modeling: The size and material information of the main microsystem structure are shown in Table III.

The system environment for chip heating test in microsystem is established. The microsystem module installed on the PCB is upturned and attached to the liquid cold plate at a constant temperature of 25 °C. The chip thermal power is 10 W. For a good contact condition between the module and the cold plate, the thermal grease is daubed between the module and the cold plate, and meanwhile, the support frame and the pressure lever are installed above the PCB. The simulation environment is shown in Fig. 7.

In order to verify the accuracy of the equivalent modeling method in this article, the detailed model and equivalent model of the microsystem are established, respectively, for thermal simulation analysis. The simulation models of the microsystem are shown in Fig. 8. The detailed local model including all microbumps is shown in Fig. 8(b). The microbump layers are equivalently treated by homogenized entities in the equivalent local model, as shown in Fig. 8(c).

To ensure the accuracy of the simulation results, the mesh independence of the detailed model is studied and the grid numbers are increased roughly by 1.5 times, respectively, grid (A)843 658, grid (B)1 234 037, grid (C)1 902 056, grid (D)2 833 083, as shown in Fig. 9. The maximum chip temperature curve is shown in Fig. 10 after simulation calculation. It can be seen that the result of grid (C) is basically stable, so the grid (C) is selected as the simulation grid model in this article. Then, final structured grid models are shown in Fig. 11, and the grid numbers of the detailed model and equivalent model are 1 902 056 and 856 627, respectively.

3) Thermal Simulation Analysis: After the simulation calculation in the Flotherm simulation software, the simulation results are as follows. The simulation temperature cloud diagrams inside the microsystem for the detailed model and the equivalent model are shown in Figs. 12 and 13. According to

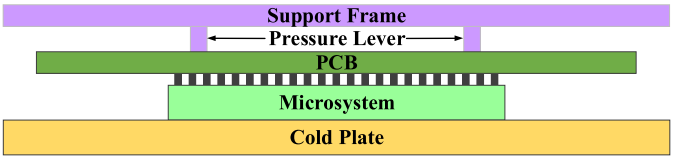


Fig. 7. Microsystem test environment schematic.

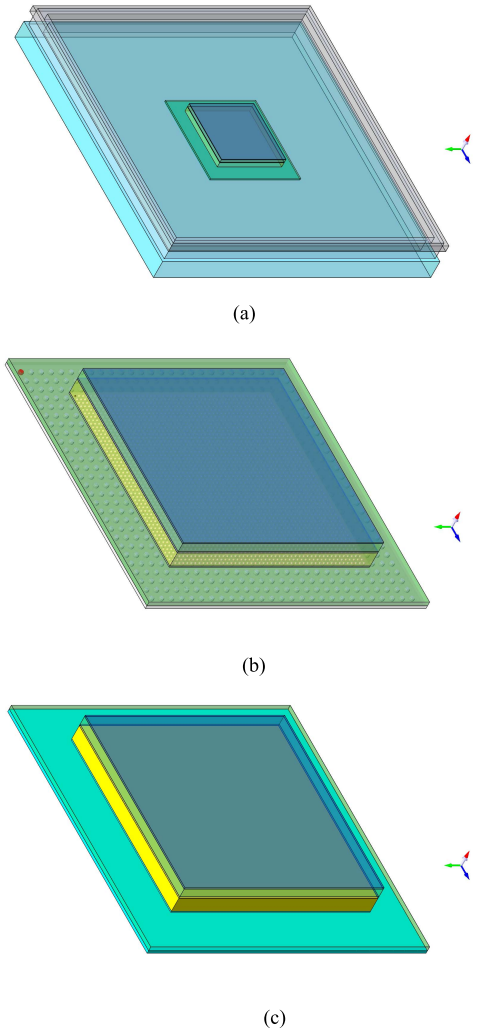


Fig. 8. Simulation models of microsystem. (a) Microsystem model. (b) Detailed local model. (c) Equivalent local model.

figures, the chip junction temperatures in the detailed model and equivalent model are 33.06 °C and 32.99 °C, respectively. In order to verify the accuracy of the equivalent modeling method, the temperature values of the microsystem on the heat dissipation path in the direction of microbump layers are extracted, including the TSV interposer temperature after the chip heat passed through the microbump layer 1 (Point 1) and the substrate temperature after the chip heat passed through the microbump layer 2 (Point 2). The comparison of the simulation results for two modeling methods is shown in Table IV.

The main conclusions according to results in Table IV are summarized in the following.

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TABLE III
SIZE AND MATERIAL INFORMATION OF THE MICROSYSTEM

No.	Structure	Size/mm	Material	$k/W{\cdot}m^{-1}{\cdot}K^{-1}$
1	Chip	10×10×0.75	Silicon	117.5
2	Thermal interface material	10×10×0.08	ME7155	11
3	Heatsink	10×10×0.4	AlN	170
4	TSV interposer	14×14×0.22	Silicon	117.5
5	Package	42×42×2.8	Alloy-42	15.8
6	Cover plate	39.9×39.9×1	Alloy-42	15.8
7	Substrate	43×43×2.85	Alumina	20
8	Column	Φ0.51×2.1	Pb90Sn10	25
9	Microbump layer 1	10×10×0.07	-	*Z*: 6.25/*XY*: 0.33
10	Microbump layer 2	14×14×0.15	-	*Z*: 8.82/*XY*: 0.37

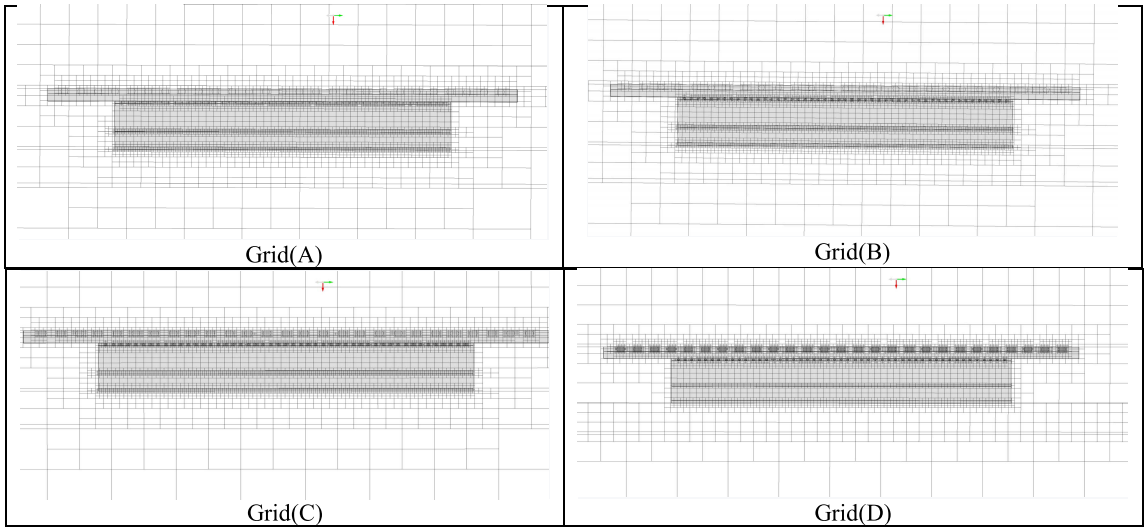


Fig. 9. Grid models of the detailed model.

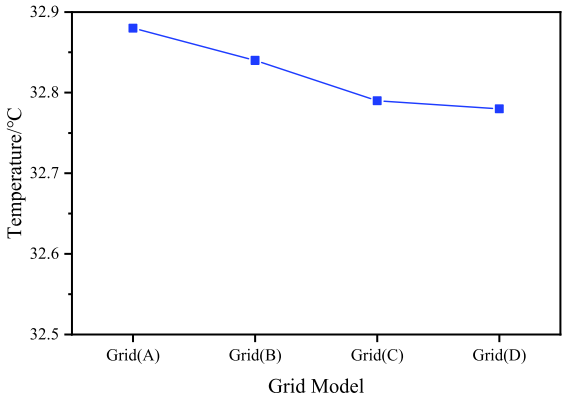


Fig. 10. Maximum chip temperature curve of different grid models.

1) The chip temperature difference calculated by the two methods is 0.07 °C, and the temperature differences at Points 1 and 2 on the heat dissipation path are 0.04 °C and 0.17 °C, respectively. The errors between two modeling methods are very small, and the calculation accuracy of the equivalent method is very high.
2) The meshing number of the detailed model is 2.22 times that of the equivalent model, while the difference of the solver time is 18.75 times. The computational efficiency of the equivalent model is significantly improved.

Therefore, the equivalent modeling method of the microbump layer adopted in this article has high simulation accuracy and greatly improves the simulation calculation efficiency.

B. Transient Thermal Test

T3ster transient tester can obtain junction temperature by monitoring the diode electrical signal of the chip. In order to accurately control the chip input power and have a diode with good temperature-sensitive characteristics, a microsystem module using the simulation temperature measurement technology of the dummy chip is manufactured. The error of the diode temperature sensor is ±0.01 °C. The simulation temperature measurement technology of dummy chip is to fabricate a thermal test chip (TTC) with heating resistance and temperature measurement unit. There are uniformly distributed heating units and temperature measurement units in this chip, and the TTC basic unit schematic is shown in Fig. 14. The heating unit can simulate the same heating power and distribution of the target chip. The microsystem chip is replaced by TTC, and other process structural forms are completely consistent with the actual module.

Fig. 15 shows the physical thermal test environment for the microsystem. The microsystem is installed upside down on PCB and then attached to the liquid cold plate, as the

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TABLE IV
SIMULATION RESULTS FOR TWO MODELING METHODS

Model	Chip temperature/℃	Point 1 temperature/℃	Point 2 temperature/℃	Meshing number	Solver time/S
Detailed model	33.06	32.79	32.38	1,902,056	3038
Equivalent model	32.99	32.83	32.55	856,627	162

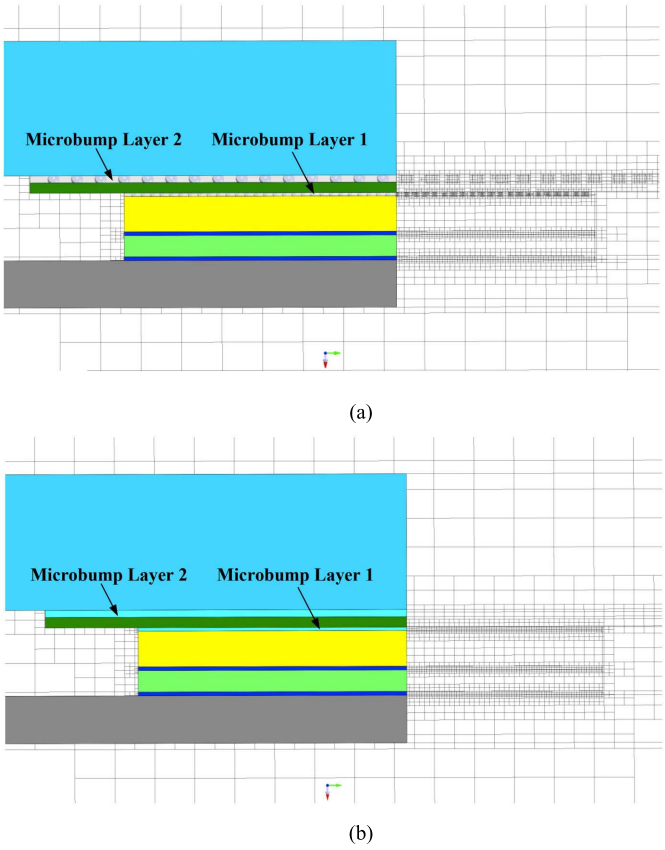


Fig. 11. Grid models of microsystem. (a) Detailed model. (b) Equivalent model.

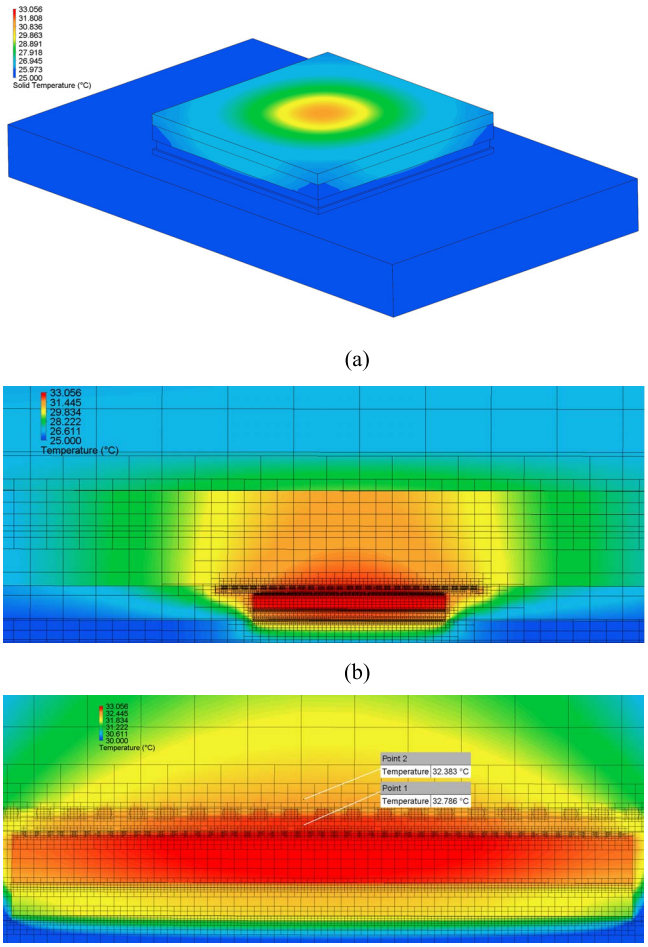


Fig. 12. Simulation temperature cloud diagram for the detailed model. (a) Global temperature cloud diagram (hide PCB). (b) Plane temperature cloud diagram. (c) Local plane temperature cloud diagram.

same setup as shown in Fig. 7, which is worth mentioning that the microsystem cannot be seen under PCB; 10-W power consumption is applied to the microsystem TTC, and the diode voltage signal of the chip is monitored by T3ster transient tester through connected wires to obtain the junction temperature change curve of the chip, as shown in Fig. 16. It can be seen that the module is approximately stable after 120 s and the steady temperature is 33.15 °C.

In order to improve the reliability of the test result, another three microsystem samples are selected for the above thermal test, and the steady-state temperatures are 33.25 °C, 32.90 °C, and 33.18 °C, respectively. The average test result of all samples is 33.12 °C. By comparing the detailed model simulation, the equivalent model simulation, and thermal test results of the microsystem, they can be seen that percentage errors of the detailed model and the equivalent model are 0.18% and 0.39%, respectively. As a result, the accuracies of different thermal simulation methods are both very high. Therefore, the equivalent modeling method for microbump layers based on the differential idea in this article has a certain application reference for the microsystem modeling and simulation.

V. COMPARISON AND DISCUSSION

According to other work [25], the combination of bumps and underfills in the microbump layer can be considered as a smear layer of composite material, shown in Fig. 17,

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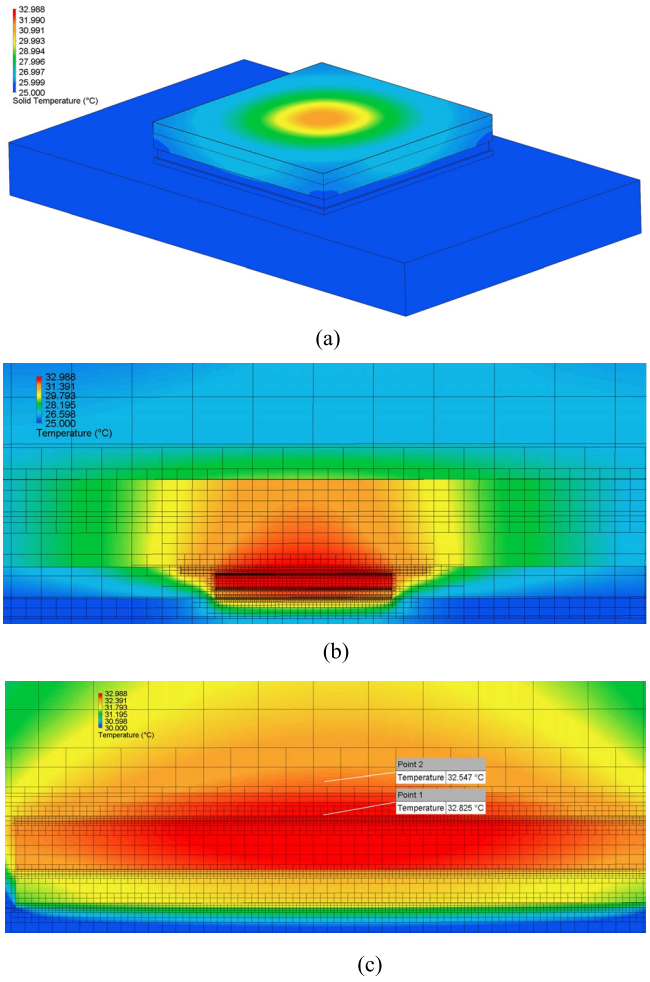


Fig. 13. Simulation temperature cloud diagram for the equivalent model. (a) Global temperature cloud diagram (hide PCB). (b) Plane temperature cloud diagram. (c) Local plane temperature cloud diagram.

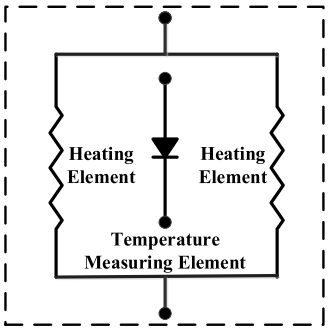


Fig. 14. TTC basic unit schematic.

where bumps are assumed to be cylindrical. Equivalent thermal conductivities in the Z- and XY-directions are defined as

$$k_y = V_b \cdot k_b + V_u \cdot k_u \tag{40}$$

$$k_z = \sqrt{k_b \cdot k_u} \cdot \frac{k_y + \sqrt{k_b \cdot k_u}}{k_c + \sqrt{k_b \cdot k_u}} \tag{41}$$

where

$$k_c = V_u \cdot k_b + V_b \cdot k_u. \tag{42}$$

TABLE V
EQUIVALENT THERMAL CHARACTERISTIC CONDUCTIVITIES OF DIFFERENT METHODS

Microbump layer	Method	$k/W{\cdot}m^{-1}{\cdot}K^{-1}$	
		Z direction	*XY* direction
Microbump layer 1	This paper	6.25	0.33
	Reference	3.92	0.86
Microbump layer 2	This paper	8.82	0.37
	Reference	6.39	1.15

The equivalent thermal conductivities of the microbump layers 1 and 2 obtained by different methods are shown in Table V.

Table V shows that the results obtained by different calculation methods vary greatly. In order to verify the difference between this article and the reference, simulation calculations under different chip powers from 10 to 40 W are carried out. The simulation results are shown in Table VI and Fig. 18.

It can be seen from the results above that: 1) the equivalent thermal conductivities in the Z-direction by the method proposed in this article are obviously bigger than the reference method results due to the actual spherical truncated crown bumps, which are taken fully into account; 2) the chip temperatures by the paper method are closer to the detailed model results; and 3) the errors compared with the detailed model increase with the increase of the chip power, and maximum errors occur at the 40-W power when the error by this article method is 0.33% and the error by the reference method is 1.61%. As a result, the equivalent calculating method based on differential idea in this article has higher accuracy.

VI. STRUCTURAL PARAMETERS ANALYSIS

The microbump layer structural parameters' influence on equivalent thermal conductivities in the microsystem is mainly studied by using the equivalent thermal conductivity calculation method based on the differential idea. The structural parameters mainly include the bump radius, the bump pitch, and the microbump layer height.

The microbump layer 2 is taken as the initial object for parameters analysis. The summary parameter dimensions are shown in Table VII, and equivalent thermal conductivities are calculated, respectively.

The calculation results of equivalent thermal conductivities in the XY-direction and Z-direction of the microbump layer under different bump radii, different bump pitches, and different microbump layer heights are shown in Figs. 19–21, respectively.

It can be seen from Fig. 19 that when the bump radius changes from 0.075 to 0.225 mm, thermal conductivities in the Z-direction continue to increase significantly, while thermal conductivities in the XY-direction increase with a small change trend. Therefore, the bump radius has a great influence on the Z-direction thermal conductivity but a little influence on the XY-direction. This is due to the proportion of the high thermal conductivity bump material in the microbump layer increases along with the increase of the bump radius, so the equivalent

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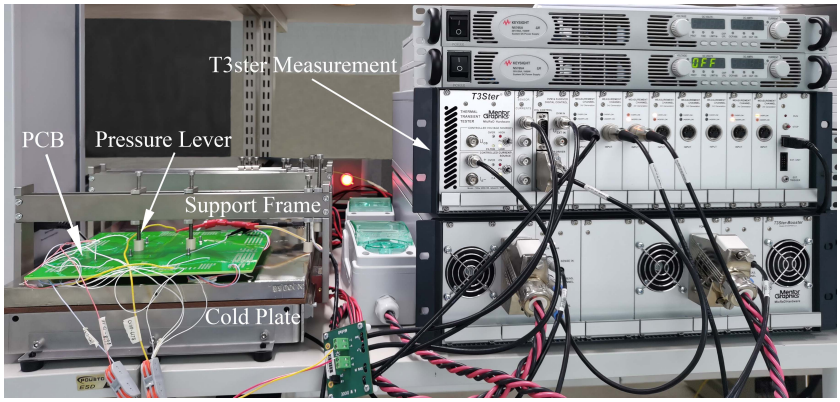


Fig. 15. Physical thermal test microsystem.

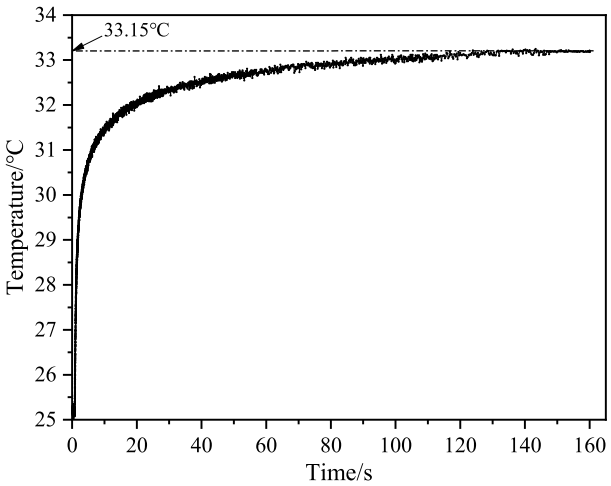


Fig. 16. Junction temperature curve of the microsystem chip.

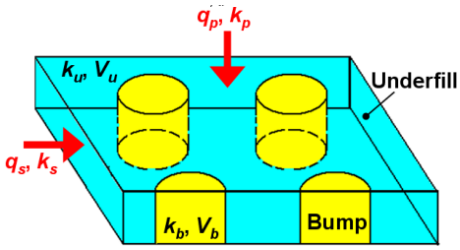


Fig. 17. Model of effective thermal conductivity for bump–underfill composite [25].

TABLE VI
SIMULATION RESULTS FOR DIFFERENT METHODS UNDER DIFFERENT CHIP POWERS

Model	Chip temperature/°C			
Power/*W*	10	20	30	40
The detailed model	33.01	40.94	48.95	57.05
This paper method model	32.99	40.89	48.89	56.86
The reference method model	33.12	41.25	49.35	57.97

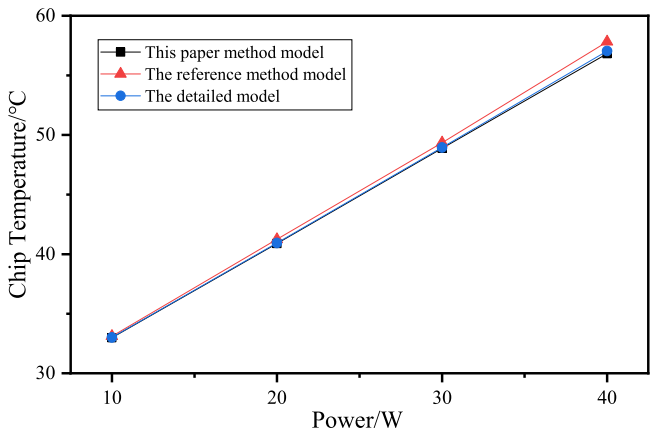


Fig. 18. Simulation results for different models.

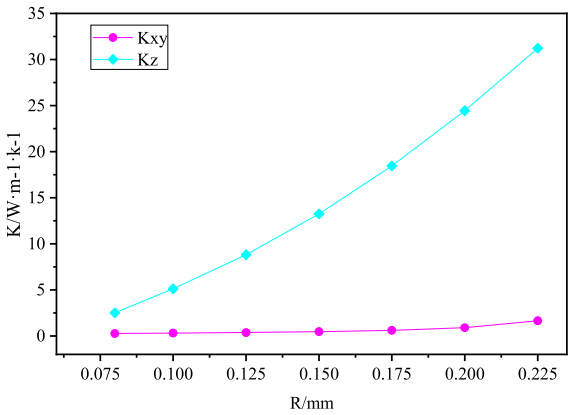


Fig. 19. Equivalent thermal conductivity results under different bump radii.

thermal conductivity in the *Z*-direction increases, while the bump radius has a little influence on the *XY*-direction due to its small value.

It can be seen from Fig. 20 that with the increase of bump pitches, the *Z*-direction thermal conductivities of the microbump layer decrease, while the overall change trend of the *XY*-direction thermal conductivity is small. Therefore, the bump pitch has a great influence on the thermal conductivity in the *Z*-direction with a decreasing trend but a little influence on the thermal conductivity in the *XY*-direction. Because the basic unit volume of the microbump layer grows and the proportion of the low thermal conductivity underfill material in the unit increases with the increase of bump pitches, so the equivalent thermal conductivity in the *Z*-direction decreases, while the bump radius still has a little influence on the *XY*-direction due to its small value.

It can be seen from Fig. 21 that the microbump layer height has a little influence on the thermal conductivity in the *Z*-direction with a slight decreasing trend but has approximately no influence in the *XY*-direction. This is because with

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TABLE VII
SUMMARY PARAMETER STRUCTURAL DIMENSIONS

Parameters	Bump radius *R/mm*	Bump pitch *p/mm*	Microbump layer height *h/mm*
Bump radius analysis	0.075/0.10/0.125/0.15 /0.175/0.20/0.225	0.50	0.15
Bump pitch analysis	0.125	0.35/0.40/0.45/0.50 /0.55/0.60/0.65	0.15
Microbump layer height analysis	0.125	0.50	0.05/0.10/0.12/0.15 /0.18/0.20/0.25

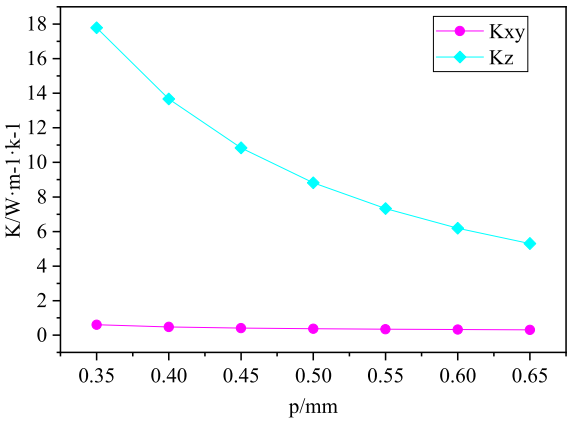


Fig. 20. Equivalent thermal conductivity results under different bump pitches.

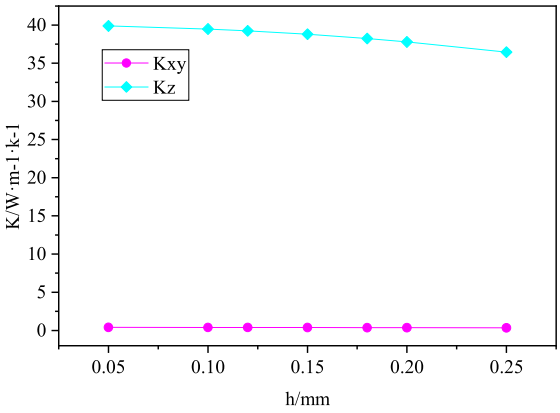


Fig. 21. Equivalent thermal conductivity results under different microbump layer heights.

the increase of the microbump layer height, the proportion of bump material in the increased part only slightly decreases, so the thermal conductivity in the Z-direction decreases a little, while the XY-direction has no influence due to the small value.

To sum up, the bump radius and the bump pitch in the microsystem have great influence on the Z-direction equivalent thermal conductivity with an increasing and decreasing trends, respectively, while the microbump layer height has a small influence on the Z-direction equivalent thermal conductivity with a decreasing trend. The three parameters all have little effect on the XY equivalent thermal conductivity. Therefore, it is necessary to focus on the influence of the bump radius and bump pitch on the microsystem thermal characteristics at the structural design stage of the microbump layer. In order to improve the overall heat dissipation ability, the bump radius can be appropriately increased and the bump pitch can be reduced on the basis of considering the process feasibility.

VII. CONCLUSION

In order to accurately simulate the actual bump structure of the microbump layer in the microsystem, the thermal simulation modeling of the microbump layer composed of spherical truncated crown bumps and underfills is studied in this article. The equivalent calculation method of thermal characteristic parameters based on the differential idea is proposed, and equivalent thermal conductivities in the Z- and XY-directions of the homogenized equivalent model are calculated, respectively. The thermal simulation analysis of the detailed model and the equivalent model is completed, and the results show that the equivalent modeling method has high accuracy and greatly calculation efficiency. Then, the accuracy of the equivalent model is also verified by comparison with the T3ster thermal transient test result and the simulation result. Also, the comparison analysis between the method in this article and the reference method is carried out. The results above show that the equivalent modeling method of the microbump layer based on the differential idea is of high accuracy, considering the actual bump structure, which provides an effective application reference for the microsystem modeling and simulation. The influence of the bump radius, the bump pitch, and the microbump layer height on heat dissipation characteristics of the microbump layer is analyzed, which provides reference for the microbump layer design in thermal management of microsystems.

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